

- `import RPi.GPIO as GPIO`
- `import time`
- `GPIO.setmode(GPIO.BCM)`
- `GPIO.setup(4, GPIO.OUT)`
- `GPIO.setup(22,GPIO.IN)`
- `while True:`
- `c = GPIO.input(22)`
- `GPIO.output(4,c)`
- `time.sleep(1)`

Command	Function
<code>GPIO.setup(22,GPIO.IN)</code>	Sets pin 22 on the Raspberry Pi as an input pin, since it will read the voltage at the switch to check if it has been pressed.
<code>c = GPIO.input(22)</code>	Reads the input on pin 22 and stores this in the variable 'c' to check if the switch has been pressed.
<code>GPIO.output(4,c)</code>	The data in variable 'c' contains the state of the switch i.e. informs whether it has been pressed or not. This command switches the LED on pin 4 ON or OFF depending on the state of the switch.

These snippets of code will look at the incremental change over the previous piece of code.

Translate with Speech Recognition and Playback

You don't have to rely on your phone to translate from one language to another anymore; you can build a translator for yourself using the Raspberry Pi.

Hardware required:

- ◆ Raspberry Pi
- ◆ USB headset

This DIY uses the Microsoft Translator service to convert speech to text,